

Predators, Prey and Population Cycles

Javalab Ecosystem Simulation · Javalab

Year 7

~30 minutes

Science · Biology

NOTICE — AI-generated worksheet

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Simulation: https://javalab.org/en/ecosystem_v2_en/

Curriculum links: AC9S7U02 · population ecology · ecosystems

Learning intentions

- I can use a population model to describe how predator and prey numbers change over time.
- I can explain why population sizes rise and fall in cycles rather than staying constant.
- I can predict the effect of changing a starting population on the whole ecosystem model.

Getting started

Open the Javalab Ecosystem Simulation. Start with the default population sizes for herbs, hares and wolves, press start, and watch the live graph build as the simulation runs.

Tasks

1. Run the simulation with the default starting populations. Describe what happens to the hare population over the first part of the graph.

2. Describe what happens to the wolf population shortly after the hare population changes. Which population seems to respond to the other, and with a delay?

3. Explain, in your own words, why the hare and wolf populations rise and fall in a repeating cycle instead of both staying steady.

4. Reset the simulation and increase the starting number of wolves. Predict what will happen to the hare population, then run it and check your prediction.

5. Try three different starting hare populations (low, medium, high) with the same wolf and herb settings. Record what happens to the hare population within the first cycle for each.

Starting hare population	What happens in the first cycle

6. Using your results, explain what might happen to the wolf population if the herb (food for hares) supply ran out completely.

7. Identify one part of this simulation that is a simplified model of a real ecosystem — something a real ecosystem has that this simulation leaves out.

Extension (optional)
Foxes and rabbits (or lynx and snowshoe hares) are a real predator-prey pair. Explain whether you would expect their real population data to show the same kind of cycling pattern you saw in the simulation, and why.
